Problem #SEPM-001: Lab Practice Problem: Comparing Static and Non-static variables in Multi-threading.

CODE:

```
import java.util.ArrayList;
import java.util.List;
import java.util.concurrent.atomic.AtomicLong;
/*
* Lab SEPM-001: Static vs Non-Static Counters in Multithreading
* NOTE: Rename the class AND the file to yourfirstname_Thread.
* (e.g. class Rahim_Thread → file Rahim_Thread.java)
*/
Public class Shahrier_Thread {
    // Experiment A: AtomicLong atomicStatic = new
                                    AtomicLong(0);
    Static long unsafeStatic = 0;

    Static class CounterTask implements Runnable {
        private final long increments;
        private final boolean threadSafe;
        long instanceCount = 0; // non-static (per object)
```

```
counterTask(long increaments boolean
    threadsafe){
        this. increments = increaments;
        this. threadsafe = threadsafe;
    }
@Override
    public void run(){}
    for(long i=0; i<increments; ++i){}
            instanceCount++;
    if(threadSafe){
        atomicStatic.incrementAndGet();
    }else{
        unsafeStatic++;
    }
  }
 }
}

// Runs one experiment returns staticCount

Non static foto
    Static long[] runExperiment(int threads
long increments, boolean threadsafe) throw
    InterruptedException{
            atomicStatic.set(0);
```

```
        unsafeStatic = 0;
    List<ThreadTask> tasks = new ArrayList<>();
    List<Thread> pool = new ArrayList<>();

    for(int i=0; i<threads; i++){

      counterTask t = new counterTask(increments,
                                      threadsafe);
          task.add(t);

        pool.add(new Thread(t));

      for(Thread t:pool) t.start();

      for(Thread t : pool) t.join();

        long nonStaticTotal = 0;

    for(counterTask t: tasks) nonStaticTotal t = t.
                                        instanceCount;
     long StaticCount = threadSafe ? atomicStatic.get():
                                        unsafeStatic;
      return new long[] {StaticCount, nonStaticTota

    static String percent(long diff, long nonstati

      if(nonStatic==0) return (diff==0)?"0.000
                                  :"undefined";

      return String.format("%.4f%%", (diff*100.
                                  nonStatic);

    }
```

```
static void report(int threads, long increments,
        boolean threadSafe) throws
InterruptedException {
long[] r = runExperiment(threads, increments,
                threadSafe);
long expected = (long) threads * increments;
    long diff = Math.abs(r[0] - r[1]);
System.out.println("Mode: " + (threadSafe?
    "ThreadSafe (AtomicLong)" : "Unsynchronized
    (long)"));
System.out.println("Threads (IV): " + threads);
System.out.println("Increments / thread (k):
    " + increments);
System.out.println("Expected (N×k): " + expected);
System.out.println("Static Count: " + r[0]);
System.out.println("Non-static total: " + r[1]);
System.out.println("Absolute difference: " + diff);
System.out.println("Percentage diff: " +
                percent(diff, r[1]);
}
```